The vulcanization system for NBR rubber composites is typically composed of a **vulcanizing agent (sulfur or peroxide)**, one or more **accelerators**, and **activators (zinc oxide and stearic acid)**. The choice of plasticizer and the curing temperature/time depend on the specific compound formulation.

□ Vulcanization System Components and Typical Proportions

| Component | Function | Typical Materials | Typical Loading (phr*) |

| :--- | :--- | :--- | :--- |

| **Vulcanizing Agent** | Forms crosslinks between rubber chains | Sulfur; Peroxides (DCP) | 0.05–5 |

| **Accelerator** | Speeds up vulcanization | Sulfenamides (CZ, CBS, DCBS), Thiazoles (MBT, MBTS), Thiurams (TMTD) | 0.2–10 total |

| **Activator** | Increases accelerator efficiency | Zinc Oxide (ZnO) + Stearic Acid | ZnO: 3–5; Stearic Acid: 0.5–1 |

| **Plasticizer** | Improves processing, lowers hardness | Phthalates, Adipates, Liquid NBR | 5–20+ |

| **Retarder** | Prevents premature scorch | CTP (N-cyclohexylthiophthalimide) | 0.25–1 |

*phr = parts per hundred rubber

□ Vulcanization Temperature and Time

The optimal cure temperature and time are determined by the **rheometer** (t90 = time to 90% cure), and they are interdependent. Higher temperatures shorten both scorch and cure times .

| Temperature | Typical t90 (Optimum Cure Time) | Typical ts2 (Scorch Time) |

| :--- | :--- | :--- |

| **150°C** | ~7–15 minutes | ~2–4 minutes |

| **160°C** | ~4–8 minutes | ~1–2 minutes |

| **170°C** | ~2–4 minutes | ~1 minute |

Example Data: For an NBR compound, increasing the temperature from 150°C to 160°C reduced ts2 from 38 seconds to 19 seconds and t90 from 684 seconds to 424 seconds .

□ Example Formulation for a Low-Temperature NBR

A patent for a low-temperature resistant NBR (usable to -45°C) provides a concrete example of a blend proportion :

* **Base**: NBR (JSRN250S): 75 phr; NBR (other grade): 4 phr; PVC: 25 phr
* **Reinforcing Filler**: 65 phr
* **Plasticizer (Softener)**: 8 phr
* **Activator**: 8.5 phr (ZnO:Stearic Acid = 2.5:5)
* **Vulcanization Accelerator**: 5 phr (a blend of DCP, DTDM, S-80, TMTD, and CZ)

□ Important Considerations

* **Plasticizer Effect**: Adding a plasticizer (e.g., 20 phr liquid rubber) can dilute the cure system. You may need to increase the sulfur and accelerator content by a similar percentage to maintain crosslink density .
* **Scorch Safety**: When mixing at high temperatures (>120°C), a single-stage mix may be unsafe. A two-stage mixing procedure or a retarder (CTP) is recommended .
* **Filler Type**: Smaller particle size carbon black (e.g., N330 vs. N774) typically shortens scorch and cure times while increasing the cure rate index .